

Interpretation of Statistical Graphs

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Statistical graphs are visual representations of data that make information easier to understand.

Bar Graph

A **bar graph** compares quantities among different categories.

For example, it can compare the number of books read by different students.

Line Graph

A **line graph** shows how data changes over time.

For example, it can show daily temperatures over several days.

Pie Chart

A **pie chart** shows how a whole is divided into parts or percentages.

The entire circle represents the whole.

Histogram

A **histogram** displays the distribution of grouped numerical data.

It shows the frequency of data within different intervals.

Interpreting Graphs

To **interpret** a graph means to read, analyze, compare, and draw conclusions from the information presented.

What to Look For

Look for the **greatest, least, trends, differences**, and possible conclusions supported by the data.

Examples

Example 1

A bar graph shows the number of books read by four students.

Student	Anna	Ben	Carlo	Diana
Books	8	5	11	7

Who read the most books?

Carlo

Example 2

Refer to the line graph of daily temperatures.

Day	Mon.	Tue.	Wed.	Thu.	Fri.
°C	30	31	33	32	34

On which day was the highest temperature recorded?

Friday

Example 3

A pie chart shows how a student spends free time.

Activity	Reading	Sports	Watching TV	Others
Percentage	20%	35%	25%	20%

Which activity occupies the largest portion?

Sports

Example 4

A histogram shows the following grouped scores.

Score	40–49	50–59	60–69	70–79	80–89
Frequency	2	4	7	9	5

Which score interval has the highest frequency?

70–79

Example 5

A bar graph shows the number of fruits sold.

Fruit	Apple	Banana	Orange	Mango
Number	18	25	20	15

How many more bananas were sold than mangoes?

$$25 - 15 = 10$$

Therefore, 10 more bananas were sold.

Example 6

Refer to the line graph.

Month	June	July	August	September
Rainy Days	12	18	21	17

During which month did the number of rainy days decrease?

September

Example 7

A pie chart shows the monthly expenses of a family.

Expense	Food	Transportation	Utilities	Others
Percentage	40%	20%	25%	15%

Which expense takes the smallest percentage?

Others

Example 8

Study the following bar graph data.

Club	Science	Math	Music	Art
Members	30	24	18	21

Which club has the second largest number of members?

Math

Practice Exercises

Practice 1

A bar graph shows the number of books read by four students.

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Books	8	5	11	7

Who read the most books?

Practice 2

Refer to the line graph of daily temperatures.

Day	Monday	Tuesday	Wednesday	Thursday	Friday
°C	30	31	33	32	34

On which day was the highest temperature recorded?

Practice 3

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Practice 5

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Refer to the line graph.

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Activities

Activity 1

A bar graph shows the number of notebooks sold.

Store	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
Notebooks	10	14	9	16

Which store sold the most notebooks?

Activity 2

Refer to the line graph of daily visitors.

Day	Monday	Tuesday	Wednesday	Thursday	Friday
Visitors	120	135	140	138	150

On which day was the greatest number of visitors recorded?

Activity 3

A pie chart shows favorite fruits.

Fruit	Apple	Banana	Mango	Orange
Percentage	30%	25%	30%	15%

Which fruit has the smallest percentage?

Activity 4

A histogram shows the following grouped ages.

Age	10–12	13–15	16–18	19–21	22–24
Frequency	5	8	11	6	3

Which age interval has the highest frequency?

Activity 5

A bar graph shows the number of bicycles rented.

Day	Monday	Tuesday	Wednesday	Thursday
Bicycles	22	18	27	20

How many more bicycles were rented on Wednesday than Tuesday?

Activity 6

Refer to the following line graph.

Week	1	2	3	4
Trees Planted	18	22	25	21

During which week did the number of trees planted decrease?

Activity 7

A pie chart shows classroom responsibilities.

Task	Cleaning	Watering Plants	Arranging C
Percentage	35%	15%	20%

Which task has the smallest percentage?

Activity 8

Study the following bar graph data.

Team	Red	Blue	Green	Yellow
Points	42	36	39	31

Which team has the second highest number of points?

Thank you!